F18CE2 - MV Calculus and Real Analysis B: Solutions 1

1 (a) We have $2a + 5b - 7L = 2(a - L) + 5(b - L)$ so

$$|2a + 5b - 7L| = |2(a - L) + 5(b - L)| \leqslant 2|a - L| + 5|b - L| < 7r$$

(b) $|3a - 2b - L| = |3(a - L) + 2(L - b)| \leqslant 3|a - L| + 2|b - L| < 5r.$

2 We have $ab - xy = b(a - x) + x(b - y)$ so

$$|ab - xy| = |b(a - x) + x(b - y)| \leqslant |b|\,|a - x| + |x|\,|b - y| < (|x| + |b|)r.$$

3 $|x^2 - 16| = |x + 4|\,|x - 4| < \delta|x + 4|\,.$

Now, $|x + 4| = |x - 4 + 8| \leqslant |x - 4| + 8 < 9 \implies |x^2 - 16| < 9\delta < r.$

4 By the triangle inequality

$$|a| = |a - b + b| \leqslant |a - b| + |b| \implies |a| - |b| \leqslant |a - b|,$$

and similarly,

$$|b| - |a| \leqslant |b - a|.$$

Combining these gives

$$-|b - a| = -|a - b| \leqslant |a| - |b| \leqslant |a - b| \implies \big||a| - |b|\big| \leqslant |a - b|.$$

5 $\lim\limits_{n\to\infty} a_n = L$ if given any $\epsilon > 0$ there exists an integer N such that

$$|a_n - L| < \epsilon \quad \text{for all } \ n \geqslant N$$

(a) Let $\epsilon > 0$. Then

$$|a_n - 5| = \frac{3}{n} < \epsilon \quad \text{if } n > \frac{3}{\epsilon}\cdot$$

Let N be an integer with $N > \frac{3}{\epsilon}\cdot$ Then

$$|a_n - 5| = \frac{3}{n} \leqslant \frac{3}{N} < \epsilon \quad \text{for all } \ n \geqslant N.$$

(b) Let $\epsilon > 0$. Then

$$\left|\frac{n + 4}{n + 6} - 1\right| = \frac{2}{n + 6} < \frac{2}{n} < \epsilon \quad \text{if } n > \frac{2}{\epsilon}\,.$$

Let N be an integer with $N > \frac{2}{\epsilon}\cdot$ Then

$$\left|\frac{n + 4}{n + 6} - 1\right| < \frac{2}{n} < \epsilon \quad \text{for all } \ n \geqslant N.$$

(c) Let $\epsilon > 0$.

$$\left|\frac{3\cos n}{n + 1} - 0\right| \leqslant \frac{3}{n + 1} \leqslant \frac{3}{n} < \epsilon \quad \text{if } n > \frac{3}{\epsilon}\cdot$$

Let N be an integer with $N > \frac{3}{\epsilon}\cdot$ Then

$$\left|\frac{3\cos n}{n + 1} - 0\right| < \frac{3}{n} < \epsilon \quad \text{for all } \ n \geqslant N.$$

6 (a) Let $\epsilon > 0$. We have,

$$|(5a_n + 3) - (5L + 3)| = |5a_n - 5L| = 5\,|a_n - L|\,.$$

Since $\lim_{n\to\infty} a_n = L$, there exists N, such that

$$n \geqslant N \implies |a_n - L| < \frac{\epsilon}{5} \implies |(5a_n + 3) - (5L + 3)| = 5\,|a_n - L| < \frac{5\epsilon}{5} = \epsilon.$$

(b) Let $\epsilon > 0$. Since $\lim_{n\to\infty} a_n = L$, there exists N, such that

$$n \geqslant N \implies |a_n - L| < \epsilon \implies \big||a_n| - |L|\big| < |a_n - L| < \epsilon.$$

7 (a) $a_n = 2 + 1/n$.

(b) $(-1)^n$.

(c) $a_n = 1$ and $b_n = (-1)^n$.

(d) $a_n = n$ and $b_n = -3n$ so that $3\,a_n + b_n = 0$.

(e) $a_n = 1/n$.

(f) If (a_n) is the sequence $0, 1, 0, 1, 0, 1, \cdots$ and (b_n) is the sequence $1, 0, 1, 0, 1, 0, \cdots$ then $a_n\,b_n = 0$ for all n, so is convergent.

(g) $a_n = 1/n$.

8 (a) Let $\epsilon > 0$. Now

$$|2a_{n+1} + 3a_n - 5L| = |(2a_{n+1} - 2L) + (3a_n - 3L)| \leqslant 2\,|a_{n+1} - L| + 3\,|a_n - L|\,.$$

Since $\lim_{n\to\infty} a_n = L$, there exists N such that, for $n \geqslant N$, $|a_n - L| < \frac{\epsilon}{5}$ and so

$$|2a_{n+1} + 3a_n - 5L| \leqslant 2\,|a_{n+1} - L| + 3\,|a_n - L| < \frac{2\epsilon}{5} + \frac{3\epsilon}{5} = \epsilon.$$

(b) Let $\epsilon > 0$. Now,

$$\begin{aligned} |a_{n+2} - 2a_{n+1} + a_n| &= |(a_{n+2} - L) + (2L - 2a_{n+1}) + (a_n - L)| \\ &\leqslant |a_{n+2} - L| + 2|a_{n+1} - L| + |a_n - L|. \end{aligned}$$

Since $\lim_{n\to\infty} a_n = L$, there exists N such that, for $n \geqslant N$, $|a_n - L| < \frac{\epsilon}{4}$, and so

$$\begin{aligned} |a_{n+2} - 2a_{n+1} + a_n| &\leqslant |a_{n+2} - L| + 2|a_{n+1} - L| + |a_n - L| \\ &< \frac{\epsilon}{4} + \frac{2\epsilon}{4} + \frac{\epsilon}{4} = \epsilon. \end{aligned}$$